

A Cross-Method Consensus Framework for Explainable AI-Based Network Intrusion Detection

Asadullah Irshad *

Centre of Excellence for Data Science, Artificial Intelligence and Modelling, University of Hull, Hull
HU6 7RX, United Kingdom

* Corresponding author. E-mail address: asadullahirshad3@gmail.com

ORCID: 0009-0005-5068-6404

Abstract

Machine learning based Intrusion Detection Systems (IDSs) are accurate but opaque, and post hoc explanation methods are typically adopted one at a time. This paper introduces a different premise: that the disagreement between explanation methods is itself a measurable signal that can be fed back into the detector's design, a closed loop not previously proposed for intrusion detection. We apply TreeSHAP, LIME, and permutation importance to Random Forest, LightGBM, and multilayer perceptron detectors, quantify how far the explanations agree with an Explanation Consensus Score, and fuse the eight resulting feature rankings by Borda counting. On NSL-KDD, UNSW-NB15, and CIC-IDS2017, agreement is low, while consensus-guided selection of ten features raises NSL-KDD F1-score from 0.766 to 0.799, performs close to or better than single-method SHAP selection on all three benchmarks, and roughly halves training time. Against mutual information, chi-square, and recursive feature elimination at the same budget, consensus selection is the only selector that improves on the full feature set under the distribution shift of NSL-KDD, with gains significant in paired tests over five runs ($p < 0.001$). All experiments use public datasets and are reproducible.

Keywords: Explainable artificial intelligence; Intrusion detection; SHAP; LIME; Explanation disagreement; Feature selection; Network security

1. Introduction

Network Intrusion Detection Systems (IDSs) are a central line of defense for modern networked infrastructures, and machine learning has become a widely used approach for building them [1, 2]. Despite long-standing warnings about the gap between laboratory accuracy and operational utility [3], deep and ensemble models now routinely reach very high detection scores on public benchmarks [4, 5]. The price of that accuracy is opacity. An analyst staring at an alert from a boosted ensemble or a neural network gets no sense of why the flow was flagged, and that gap hurts everything downstream: triage, incident response, forensic write-ups, regulatory accountability [6-9]. Explainable Artificial Intelligence (XAI) has therefore become a major research direction for security analytics in operational environments, where large volumes of network flows must be classified and, when flagged, justified [10].

The de facto practice in explainable intrusion detection is to attach a single post hoc explanation method, most commonly SHAP [12] or LIME [11], to a trained detector and to present the resulting feature attributions as the explanation of the model. Baked into this habit are two assumptions: that the chosen explainer is faithful, and that another reasonable explainer would have told much the same story. Recent work on the disagreement problem in explainable machine learning shows that this assumption is fragile in general [14], and that post hoc explainers can even be manipulated adversarially [15]. For intrusion detection specifically, the assumption is largely untested at scale: existing studies either use one explainer, or compare explainers qualitatively on a single dataset and model.

This paper takes that untested assumption seriously. It measures explanation agreement for intrusion detection systematically and then, rather than treating the disagreement as bad news, puts it to work. We apply three widely used attribution methods, TreeSHAP [12, 34], LIME [11], and permutation importance [13], to three detector families, Random Forest, LightGBM, and a multilayer perceptron (MLP), trained on three public benchmarks, NSL-KDD, UNSW-NB15, and CIC-IDS2017. From the resulting grid of feature rankings we compute pairwise agreement statistics and summarize them in a single Explanation Consensus Score (ECS). All eight rankings are then fused with a Borda count, and the fused ranking drives feature selection, so that the compact detector we end up with rests on features that receive consistent support across the evaluated explainer-model pipelines, rather than on the verdict of a single method.

The novelty of this work is conceptual before it is technical. Prior explainable-IDS studies treat a single post hoc explainer as the explanation of a detector; the few that compare explainers do so qualitatively, on one model and one dataset, and stop at observing that the explanations differ. We instead make explanation disagreement a first-class, quantitative object, measure it systematically across a full grid of methods, models and datasets, and then, uniquely, feed the resulting consensus back into the detection pipeline as a feature-selection principle. The contribution is thus not a new explainer or a new detector, but a new use of explanations, turning an agreement audit into measurable gains in generalization, compactness and training cost.

The paper makes five contributions. First, we provide a systematic measurement of explanation agreement across eight explainer-model combinations per dataset over three public IDS benchmarks. Second, we introduce ECS, a deliberately simple agreement metric built on Jaccard overlap of top-ranked features, backed by Kendall rank correlations over the full rankings. Third, we show that agreement is low, with ECS between 0.213 and 0.340, meaning that two arbitrarily chosen explanation pipelines share on average only about a fifth to a third of their top-10 features, which has important implications for how XAI evidence is interpreted and reported in security operations. Fourth, we propose Borda-consensus feature selection and evaluate it against four standard selectors: SHAP-only importance, mutual information, chi-square, and recursive feature elimination. With only ten consensus features, F1-score on NSL-KDD improves from 0.766 to 0.799 relative to the full 122-feature model, and consensus selection outperforms every baseline on NSL-KDD with margins of 4.0 to 9.4 F1 points, all statistically significant in paired tests over five repeated runs ($p < 0.001$), while matching the baselines on the other two benchmarks at roughly half the training cost of the full models. Fifth, the complete measurement protocol is released, so every number, table, and figure here can be regenerated from the public datasets with fixed seeds.

Section 2 reviews related work; Section 3 describes the datasets, models, explanation methods, agreement metrics, and the consensus framework; Section 4 reports the results; Section 5 discusses implications and limitations; Section 6 concludes.

2. Related work

2.1. Machine learning for intrusion detection

Surveys of machine learning based intrusion detection document a steady progression from classical shallow models to deep architectures [1, 2, 4, 5]. Sommer and Paxson [3] famously cautioned that the semantic gap between detection scores and actionable operator knowledge limits real-world adoption, an argument that anticipates the present interest in explainability. Benchmark datasets have played a decisive role in this literature: NSL-KDD [28] remains the most widely used legacy benchmark and is deliberately constructed so that its official test partition contains attack variants absent from training;

UNSW-NB15 [29] and CIC-IDS2017 [30] provide more contemporary traffic and attack mixes. Yang et al. [31] published a carefully sampled version of CIC-IDS2017 in connection with their multi-tiered hybrid IDS, which we adopt for tractability and comparability.

2.2. Post hoc explanation methods

LIME [11] fits a sparse local surrogate model around a perturbed neighborhood of the instance being explained. SHAP [12] unifies several attribution schemes under Shapley value theory, and its TreeSHAP variant [34] computes exact or path-dependent attributions for tree ensembles efficiently. Permutation importance, formalized as model reliance by Fisher et al. [13] following Breiman's original proposal for random forests [32], measures the performance drop caused by permuting one feature at a time and is model-agnostic and global by construction. Broader taxonomies of explanation methods are given in surveys [7, 8, 24, 27], foundational methodological discussions appear in Refs. [25, 26], and Rudin [6] argues that post hoc explanation of black boxes should be replaced by inherently interpretable models in high-stakes settings, a position that makes quantitative audits of post hoc explainers, such as ours, all the more necessary.

2.3. Explainable AI for intrusion detection

Wang et al. [16] provided an influential SHAP-based framework for interpreting IDS decisions on NSL-KDD. Warnecke et al. [17] evaluated explanation methods for security-oriented deep models along fidelity, sparsity, stability, and efficiency axes, focusing on deep architectures and malware or vulnerability discovery tasks. Zebin et al. [21] built an explainable balanced-stacking IDS for DNS-over-HTTPS attacks, and Mane and Rao [22] demonstrated a multi-explainer pipeline on NSL-KDD without quantifying inter-explainer agreement. More recently, Arreche et al. [18] proposed E-XAI, an end-to-end evaluation of black-box explanation frameworks for network intrusion detection; Gaspar et al. [19] studied LIME and SHAP applicability to MLP-based IDS; and Hermosilla et al. [20] compared SHAP and LIME on UNSW-NB15 from a forensic-readiness perspective. Surveys of XAI in cybersecurity [10, 23] confirm that most published systems adopt a single explainer.

What separates the present work from this literature comes down to two things. First, prior IDS studies have generally evaluated explainers on narrower combinations of models and datasets, often without systematically quantifying cross-method agreement, whereas we quantify agreement across a full grid of three methods, three models, and three datasets, using rank statistics rather than visual inspection, which amounts to instantiating the disagreement problem [14] in the intrusion detection domain. Second, to our knowledge, prior IDS studies have not systematically used cross-method explanation agreement across multiple explainers and detector families as a consensus signal for feature selection; our Borda-consensus feature selection turns the agreement audit into measurable gains in accuracy, compactness, and training cost.

Table 1 positions this study against the closest prior work along the dimensions that define our contribution. Two gaps are visible at a glance. First, no prior IDS study quantifies cross-method explanation agreement with rank statistics: comparisons, where they exist, are qualitative and limited to two explainers on a single model or dataset. Second, no prior study closes the loop from multi-explainer evidence back into the detection pipeline; explanations are reported, not used. This paper fills both gaps.

Table 1. Positioning of this study against the closest prior work on explainable intrusion detection.

Study	Explanation methods	Detector families	IDS datasets	Agreement quantified	Consensus fed back
Wang et al. [16]	SHAP	1	1	No	No

Study	Explanation methods	Detector families	IDS datasets	Agreement quantified	Consensus fed back
Warnecke et al. [17]	6 (deep models)	Deep only	Security tasks	Criteria-based, no rank agreement	No
Mane and Rao [22]	SHAP, LIME, others	1	1	No	No
Arreche et al. [18]	SHAP, LIME	Several	3	Qualitative	No
Gaspar et al. [19]	SHAP, LIME	MLP only	1	Qualitative	No
Hermosilla et al. [20]	SHAP, LIME	Several	1	Qualitative	No
This work	SHAP, LIME, PI (8 pipelines)	3 (bagging, boosting, neural)	3	Jaccard@k, Kendall τ , ECS	Yes (Borda $\rightarrow$ feature selection)

3. Methods

Fig. 1 sketches the framework. Three benchmarks feed three detector families; three attribution methods are applied to each trained detector; the resulting rankings are compared with agreement statistics, fused by Borda counting, and the fused ranking drives feature selection for a compact final detector. The pieces are described in turn below.

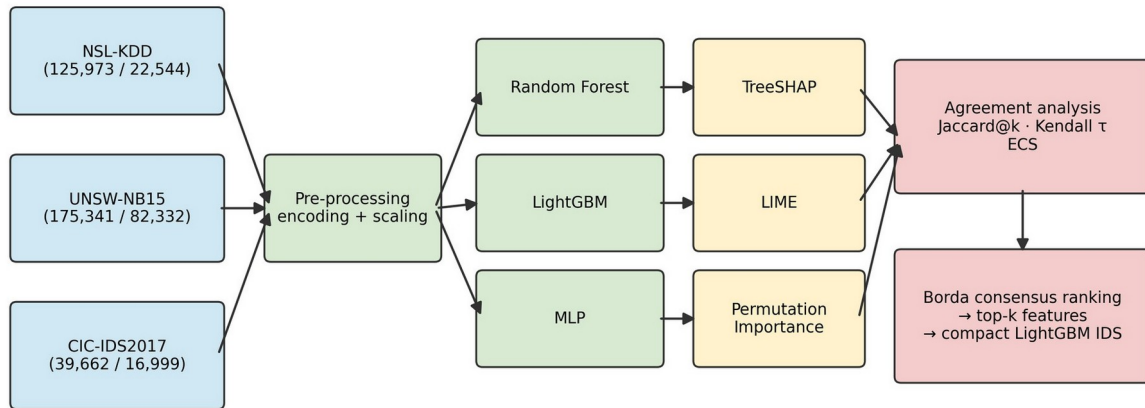


Fig. 1. Overview of the cross-method consensus framework for explainable intrusion detection.

3.1. Datasets and preprocessing

NSL-KDD [28] is used with its official KDDTrain+ and KDDTest+ partitions (125 973 and 22 544 records). The 41 features include three categorical fields (protocol type, service, and flag) which are one-hot encoded, yielding 122 input features; labels are binarized to normal versus attack. The official test partition contains attack types absent from training, which makes NSL-KDD a de facto test of generalization to unseen attacks and explains why published accuracies on KDDTest+ cluster far below the near-perfect values obtained with random splits of the same data.

UNSW-NB15 [29] is used with its official training and testing partitions (175 341 and 82 332 records) and 42 features after removing the record identifier; the categorical protocol, service, and state fields are integer-coded in the public partitioned release we use. CIC-IDS2017 [30] is used through the published sample of Yang et al. [31] (56 661 records over 77 numeric flow features), split 70/30 with stratification; labels are binarized to benign versus attack, infinite values are replaced and missing

values imputed with zero. All features on all datasets are standardized with statistics computed on the training partition only. Table 2 summarizes the resulting corpora.

Table 2. Benchmark datasets after preprocessing.

Dataset	Training records	Test records	Features	Attack ratio (train/test)
NSL-KDD	125,973	22,544	122	46.5% / 56.9%
UNSW-NB15	175,341	82,332	42	68.1% / 55.1%
CIC-IDS2017	39,662	16,999	77	59.9% / 59.9%

3.2. Detection models

Four classifiers are trained per dataset with fixed random seed 42: logistic regression (LR) as a linear baseline; Random Forest (RF) [32] with 100 trees; LightGBM [33] with 200 boosting rounds and 63 leaves; and an MLP with two hidden layers of 64 and 32 units, early stopping, and at most 40 epochs. LR serves only as a performance reference; the explanation study uses RF, LightGBM, and the MLP, which represent the bagging, boosting, and neural model families that dominate the applied IDS literature. Performance is reported as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC) on the held-out test partition, together with training time and per-flow inference latency measured on the same commodity CPU environment.

3.3. Explanation methods

Three post hoc attribution methods are applied. (1) SHAP: TreeSHAP [34] is applied to RF and LightGBM on 2000 randomly drawn test flows (1000 with the fast approximation for RF); global importance is the mean absolute SHAP value per feature. (2) LIME [11]: local surrogate explanations with 1000 perturbations are computed for 50 randomly drawn test flows per model, for all three models, and aggregated into a global ranking by mean absolute local weight. (3) Permutation importance (PI) [13, 32]: computed for all three models on 3000 test flows with five repetitions. This yields eight explainer-model rankings per dataset: SHAP on two models, and LIME and PI on three models each. Kernel-based SHAP for the MLP was excluded because its cost is prohibitive at IDS scale, an operational constraint discussed in Section 4.6.

3.4. Agreement metrics and the Explanation Consensus Score

Let r and s be two rankings of the d features by decreasing importance, and let $T(r, k)$ denote the set of the k top-ranked features of r . Agreement at depth k is measured by the Jaccard index $J(r, s; k) = |T(r, k) \cap T(s, k)| / |T(r, k) \cup T(s, k)|$, and agreement over full rankings by Kendall's rank correlation coefficient τ [35]. With m available rankings, the Explanation Consensus Score of a dataset is defined as the mean pairwise Jaccard index at depth $k = 10$ over all $m(m - 1)/2$ pairs. ECS lies in $[0, 1]$; the value expected under independent random top-10 sets is approximately $k/(2d - k)$, i.e., about 0.04 for NSL-KDD ($d = 122$), 0.07 for CIC-IDS2017 ($d = 77$), and 0.14 for UNSW-NB15 ($d = 42$). We also report intra-model agreement (same model, different explainers) and intra-method agreement (same explainer, different models) separately.

3.5. Borda consensus and explanation-guided feature selection

The eight rankings are fused with a Borda count: a feature ranked at position i in a ranking over d features receives $d - i$ points, and features are re-ranked by total points. The top- k consensus features are then used to retrain a LightGBM detector for $k \in \{5, 10, 15, 20, 30\}$, compared against the full feature set and against a SHAP-only baseline that selects the top ten features from the SHAP ranking

of the LightGBM detector. Borda counting is used because it requires no calibration between heterogeneous importance scales, only ranks, and because it degrades gracefully when one explainer is unreliable.

Algorithm 1. Cross-method consensus framework.

```

Input: dataset  $D = (X_{tr}, y_{tr}, X_{te}, y_{te})$  with  $d$  features; model set  $M = \{RF, \text{LightGBM}, \text{MLP}\}$ ;
explainer set  $E = \{\text{TreeSHAP}, \text{LIME}, \text{PI}\}$ ; depth  $k$ ; budget  $k'$ 
Output: consensus ranking  $\rho^*$ ; compact detector  $f^*$ ; consensus score ECS
1: for each  $m \in M$ : train  $m$  on  $(X_{tr}, y_{tr})$ 
2:  $R \leftarrow \emptyset$   $\triangleright$  set of feature rankings
3: for each  $(m, e) \in M \times E$  with  $e$  applicable to  $m$ :
4:    $\rho(m, e) \leftarrow$  rank features by global attribution of  $e$  on  $m$ ;  $R \leftarrow R \cup \{\rho(m, e)\}$ 
5:  $\text{ECS} \leftarrow$  mean over pairs  $\{r, s\} \subseteq R$  of  $J(r, s; k)$   $\triangleright$  Jaccard overlap of top- $k$  sets
6: for each feature  $j$  and ranking  $r \in R$ :  $\text{score}(j) += d - \text{rank}_r(j)$   $\triangleright$  Borda count
7:  $\rho^* \leftarrow$  features sorted by descending score
8:  $f^* \leftarrow$  retrain LightGBM on the top- $k'$  features of  $\rho^*$ 
9: return  $\rho^*, f^*, \text{ECS}$ 

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Complexity. Let n be the number of training flows, d the number of features, and $m = |R|$ the number of explainer-model pipelines ($m = 8$ here). Detector training costs are the standard costs of the underlying learners. On top of training, TreeSHAP is $O(\text{TLD}^2)$ per explained instance for an ensemble of T trees of maximum depth D with L leaves; LIME costs $O(p)$ surrogate evaluations per instance for p perturbations ($p = 1000$ here); and permutation importance costs $O(r \cdot d)$ model evaluations for r repetitions. The consensus stage itself is negligible: the Borda fusion is $O(m \cdot d + d \log d)$ and the ECS computation is $O(m^2 \cdot k)$. The framework therefore adds essentially no overhead beyond the explanation methods it audits, which is what allows it to be dropped into an existing XAI-IDS pipeline.

3.6. Baseline selectors and statistical protocol

Consensus-guided selection is compared against four standard selectors at the same feature budget. (1) SHAP-only: the top- k features by mean absolute TreeSHAP value of the LightGBM detector, the single most common pipeline in the XAI-IDS literature. (2) Mutual information: the top- k features by mutual information with the class label, estimated on a 20 000-flow stratified subsample of the training partition. (3) Chi-square: the top- k features by the χ^2 statistic computed on min-max scaled training data. (4) Recursive feature elimination (RFE): backward elimination with a LightGBM estimator, removing 10% of the remaining features per step until k remain. All selectors see only the training partition.

Statistical protocol. Because LightGBM training is deterministic for a fixed dataset, robustness is assessed by refitting every configuration on five stratified 90% subsamples of the training partition (fixed seeds), always evaluating on the untouched official test partition. Headline tables report the full-training-partition run; the comparison in Section 4.5 reports mean $\pm$ standard deviation over the five runs, with paired t-tests between consensus selection and each baseline on per-run F1-scores. Table 3 lists every hyperparameter of the pipeline; no per-dataset tuning is performed anywhere.

Table 3. Pipeline hyperparameters (identical across datasets; no per-dataset tuning).

Component	Setting
Logistic regression	max_iter = 300, seed 42 (performance reference only)
Random Forest	100 trees, seed 42
LightGBM	200 boosting rounds, 63 leaves, seed 42
MLP	hidden layers 64/32, early stopping, max 40 epochs, seed 42

Component	Setting
TreeSHAP	2000 random test flows (1000 with fast approximation for RF)
LIME	50 random test flows per model, 1000 perturbations each
Permutation importance	3000 test flows, 5 repetitions
Mutual information	20 000-flow stratified training subsample, seed 42
Chi-square	min-max scaled training partition
RFE	LightGBM estimator, step = 10% of remaining features, target k = 10
Consensus	Borda count over all 8 rankings; ECS at depth k = 10
Robustness runs	5 stratified 90% training subsamples, fixed seeds; paired t-tests on F1

4. Results

4.1. Detection performance

Table 4 reports detection performance for all twelve dataset-model combinations. LightGBM and RF are the strongest detectors overall: on CIC-IDS2017 LightGBM reaches 99.80% accuracy and 0.9998 AUC, and on UNSW-NB15 87.20% accuracy and 0.9837 AUC, in line with published results for gradient-boosted ensembles on these benchmarks. On the deliberately hard KDDTest+ partition, accuracies fall in the 75%-81% band with AUC up to 0.963, reproducing the well-documented generalization gap of NSL-KDD to unseen attack variants rather than the inflated near-perfect figures obtained with random resplits. Inference stays under 7 microseconds per flow for every model, which is comfortably within budget for high-throughput pipelines.

Table 4. Detection performance on the held-out test partitions (seed 42, identical hardware).

Dataset	Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)	Train (s)	Latency (μ s/flow)
NSL-KDD	LR	75.42	91.75	62.43	74.31	78.15	5.81	1.0
NSL-KDD	RF	76.48	96.70	60.75	74.62	95.94	4.53	4.3
NSL-KDD	LightGBM	77.94	96.67	63.43	76.60	96.28	2.25	6.5
NSL-KDD	MLP	80.97	92.58	72.36	81.23	91.36	6.50	0.4
UNSW-NB15	LR	80.89	75.12	97.62	84.91	94.85	4.60	0.2
UNSW-NB15	RF	87.22	81.94	98.49	89.46	97.81	13.18	3.7
UNSW-NB15	LightGBM	87.20	82.13	98.11	89.41	98.37	2.56	3.9
UNSW-NB15	MLP	85.86	80.87	97.34	88.35	97.48	11.65	0.3
CIC-IDS2017	LR	90.42	89.64	94.97	92.23	95.59	2.45	0.3
CIC-IDS2017	RF	99.54	99.62	99.62	99.62	99.94	2.53	4.4
CIC-IDS2017	LightGBM	99.80	99.78	99.88	99.83	99.98	2.08	6.4
CIC-IDS2017	MLP	95.64	94.68	98.24	96.43	99.13	1.95	0.4

4.2. Global explanations

Fig. 2 shows the fifteen most important features by mean absolute SHAP value for the LightGBM detector on each dataset. The attributions make sense from a security standpoint: on NSL-KDD, traffic volume (src_bytes, dst_bytes), connection counts, and specific services dominate; on UNSW-NB15, time-to-live derived features (sttl), connection-state statistics (ct_state_ttl), and byte statistics lead, consistent with prior SHAP analyses of this benchmark; on CIC-IDS2017, TCP window initialization bytes and backward packet-length statistics dominate, which is characteristic of the DoS and port-scan

heavy attack mix. Fig. 3 presents the SHAP beeswarm for UNSW-NB15, showing the direction of each effect at the level of individual flows.

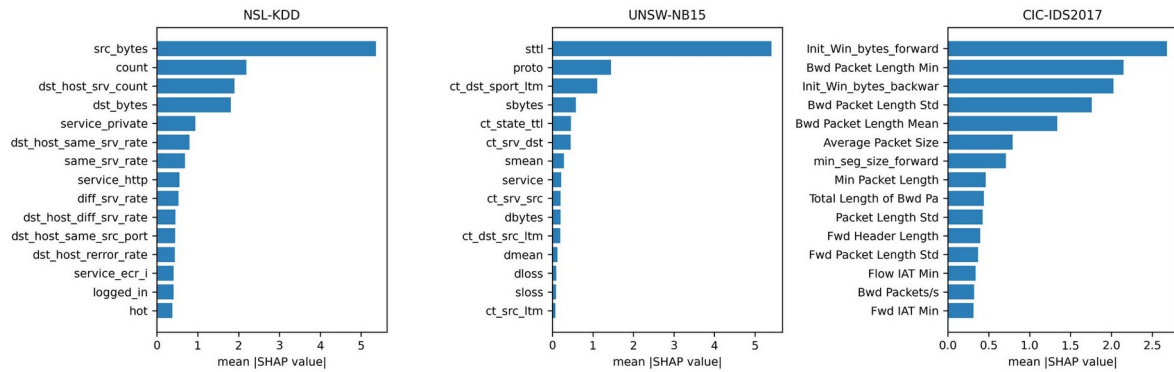


Fig. 2. Global SHAP feature importance (top 15) of the LightGBM detector on the three benchmarks.

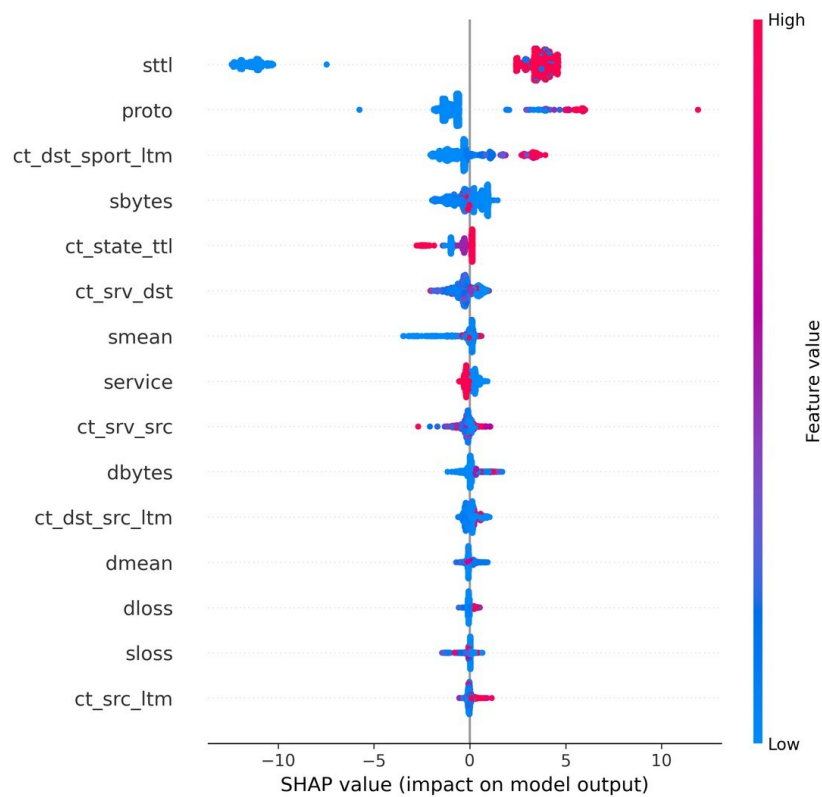


Fig. 3. SHAP beeswarm plot for LightGBM on UNSW-NB15 (2000 test flows).

4.3. How much do explanations agree?

Fig. 4 shows the pairwise Jaccard overlap of top-10 feature sets for all eight explainer-model pipelines on each dataset, and Table 5 summarizes the agreement statistics. The headline result is blunt: ECS is only 0.213 on NSL-KDD, 0.340 on UNSW-NB15, and 0.276 on CIC-IDS2017. Although these values are far above the random-overlap baselines derived in Section 3.4 (0.04-0.14), they mean that two arbitrarily chosen explanation pipelines share, on average, only about three to five of their ten top-ranked features. Kendall correlations over full rankings show the same pattern. Decomposing the score reveals that fixing the model and varying the explainer (intra-model agreement, 0.248-0.367) yields roughly as much disagreement as fixing the explainer and varying the model (intra-method agreement, 0.237-0.407); the choice of explanation method alone can change the reported evidence as much as

changing the underlying detector family. Agreement is highest on UNSW-NB15, where a few features such as sttl are so dominant that all pipelines recover them, and lowest on NSL-KDD, whose 122 sparse one-hot features admit many near-equivalent explanations.

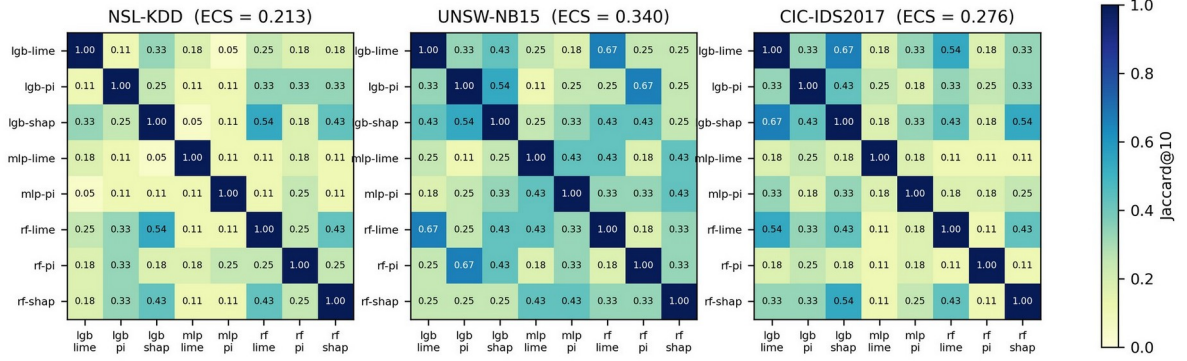


Fig. 4. Pairwise Jaccard@10 agreement between the eight explainer-model pipelines on each benchmark.

Table 5. Explanation agreement statistics per dataset.

Dataset	ECS (Jaccard@10)	Intra-model mean	Intra-method mean
NSL-KDD	0.213	0.248	0.237
UNSW-NB15	0.340	0.367	0.407
CIC-IDS2017	0.276	0.322	0.281

For security operations the warning is concrete. A report that presents SHAP values from one model as the explanation of an alert suggests a uniqueness that is not there; the observed Jaccard scores indicate that a substantial proportion of the top-ranked features can differ across explanation pipelines.

4.4. Consensus-guided feature selection

Table 6 and Fig. 5 evaluate the Borda-consensus ranking as a feature selector. Three things stand out. First, on NSL-KDD the ten-feature consensus detector outperforms the full 122-feature detector by a wide margin, 80.54% versus 77.94% accuracy and 0.799 versus 0.766 F1-score, indicating that consensus selection can act as a regularizer by prioritizing features that receive more consistent support across the evaluated explanation pipelines, and thereby improves generalization to the unseen attacks of KDDTest+. Second, against the standard SHAP-only baseline at the same budget of ten features, consensus selection is decisively better on NSL-KDD (80.54% vs. 76.52% accuracy, a 4.0-point margin) and produces very similar results to SHAP-only selection on UNSW-NB15 (86.63% vs. 87.12%) and CIC-IDS2017 (99.58% vs. 99.59%); at $k = 15$ the consensus detector surpasses both the SHAP-only baseline and the full feature set on UNSW-NB15 (87.71% accuracy, 0.8982 F1-score) while still using barely a third of the original features. In short, consensus selection incurs only small performance differences relative to the best single-method baseline while substantially reducing the feature set and training cost. Third, compactness pays for itself: with 10-15 consensus features, training time roughly halves on every dataset and AUC stays within 0.4 points of the full model.

Table 6. LightGBM performance under explanation-guided feature selection.

Dataset	Feature set	k	Acc. (%)	F1 (%)	AUC (%)	Train (s)
NSL-KDD	All features	122	77.94	76.60	96.28	2.02
NSL-KDD	Consensus top-5	5	72.05	68.33	74.57	0.73
NSL-KDD	Consensus top-10	10	80.54	79.86	96.62	1.02
NSL-KDD	Consensus top-15	15	78.97	77.91	97.28	1.22

Dataset	Feature set	k	Acc. (%)	F1 (%)	AUC (%)	Train (s)
NSL-KDD	Consensus top-20	20	78.64	77.48	97.22	1.21
NSL-KDD	Consensus top-30	30	78.14	76.85	96.95	1.50
NSL-KDD	SHAP-only top-10	10	76.52	74.68	94.98	0.87
UNSW-NB15	All features	42	87.20	89.41	98.37	2.52
UNSW-NB15	Consensus top-5	5	86.45	88.71	97.51	0.82
UNSW-NB15	Consensus top-10	10	86.63	88.90	97.93	1.19
UNSW-NB15	Consensus top-15	15	87.71	89.82	98.50	1.36
UNSW-NB15	Consensus top-20	20	87.43	89.57	98.42	1.46
UNSW-NB15	Consensus top-30	30	87.26	89.46	98.39	1.93
UNSW-NB15	SHAP-only top-10	10	87.12	89.21	97.84	1.01
CIC-IDS2017	All features	77	99.80	99.83	99.98	2.06
CIC-IDS2017	Consensus top-5	5	98.34	98.63	99.37	0.32
CIC-IDS2017	Consensus top-10	10	99.58	99.65	99.93	0.49
CIC-IDS2017	Consensus top-15	15	99.70	99.75	99.96	0.65
CIC-IDS2017	Consensus top-20	20	99.71	99.75	99.97	0.70
CIC-IDS2017	Consensus top-30	30	99.80	99.83	99.98	0.96
CIC-IDS2017	SHAP-only top-10	10	99.59	99.66	99.93	0.44

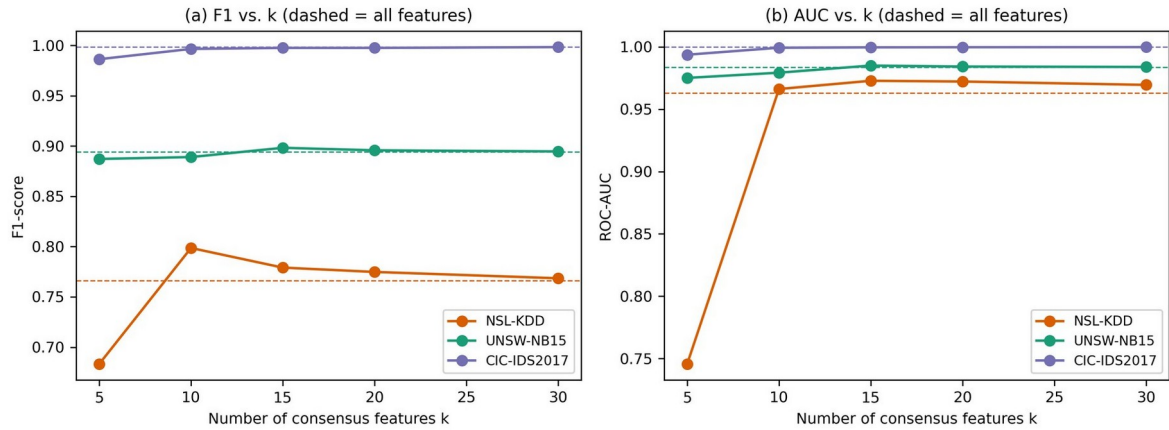


Fig. 5. F1-score (a) and AUC (b) of the consensus-k detector versus k; dashed lines mark the all-features baseline.

4.5. Comparison with standard feature-selection baselines

The editorially decisive question for any new selection method is whether it beats the selectors practitioners already use. Table 7 compares consensus selection at $k = 10$ and $k = 15$ against SHAP-only, mutual information, chi-square, and RFE at the matched budget $k = 10$, under the five-run protocol of Section 3.6. Three findings stand out.

First, under distribution shift the consensus is the only selector that helps. On NSL-KDD, whose official test partition contains attack families absent from training, consensus@10 reaches 0.794 ± 0.008 F1 and outperforms every baseline by wide, statistically significant margins: +5.0 points over SHAP-only, +5.8 over mutual information, +9.4 over chi-square, +4.0 over RFE, and +2.2 over the full 122-feature model (paired t-tests, all $p \leq 0.003$, most $p < 0.001$). Every single-method selector is worse than simply keeping all features; only the consensus improves on it.

Second, no single baseline is dependable across datasets. Chi-square collapses on NSL-KDD (0.700 F1, AUC 0.803) and loses 2.7 to 4.0 points everywhere else; mutual information is competitive on UNSW-NB15 but weak on NSL-KDD and CIC-IDS2017; RFE is strong on CIC-IDS2017 yet mediocre on NSL-KDD, and it is also the most expensive selector, requiring a sequence of model refits. The consensus never trails the best baseline by more than 0.7 F1 points on any dataset, and its top-10 sets overlap the single-method selections only partially (3 to 8 features of 10), confirming that it is not reducible to any one of them.

Third, the consensus budget can be tuned with the same machinery: at $k = 15$ the consensus detector significantly exceeds the full 42-feature model on UNSW-NB15 (0.8972 ± 0.0013 vs. 0.8953 ± 0.0009 , paired $p = 0.005$) while using about a third of the features, and sits within 0.1 points of the full model on CIC-IDS2017. Together with the ECS audit, this yields a defensible, reproducible recipe: audit agreement, fuse by Borda count, and select at a budget validated on the training side.

Table 7. Consensus selection vs. standard selectors: F1 (mean $\pm$ std over five stratified 90% training subsamples, evaluated on the official test partitions), AUC, and training time. Best selector per dataset in each budget class is discussed in the text; $\dagger$ marks configurations significantly below consensus@10 (paired t-test, $p < 0.01$).

Dataset	Selector	F1 (mean $\pm$ std)	Δ F1 vs cons@10	AUC	Train (s)
NSL-KDD	All features (122)	$0.7726 \pm 0.0086 \dagger$	-0.0216	0.9629	2.18
NSL-KDD	Consensus@10	0.7942 ± 0.0075	—	0.9637	1.01
NSL-KDD	Consensus@15	$0.7742 \pm 0.0082 \dagger$	-0.0200	0.9718	1.30
NSL-KDD	SHAP-only@10	$0.7445 \pm 0.0035 \dagger$	-0.0497	0.9521	1.01
NSL-KDD	Mutual info@10	$0.7365 \pm 0.0092 \dagger$	-0.0577	0.9507	1.03
NSL-KDD	Chi-square@10	$0.7000 \pm 0.0032 \dagger$	-0.0943	0.8033	0.88
NSL-KDD	RFE@10	$0.7546 \pm 0.0083 \dagger$	-0.0396	0.9607	1.12
UNSW-NB15	All features (42)	0.8953 ± 0.0009	+0.0069	0.9843	2.75
UNSW-NB15	Consensus@10	0.8884 ± 0.0009	—	0.9790	1.19
UNSW-NB15	Consensus@15	0.8972 ± 0.0013	+0.0088	0.9845	1.40
UNSW-NB15	SHAP-only@10	0.8956 ± 0.0009	+0.0072	0.9826	1.15
UNSW-NB15	Mutual info@10	0.8936 ± 0.0008	+0.0053	0.9802	1.22
UNSW-NB15	Chi-square@10	$0.8615 \pm 0.0003 \dagger$	-0.0269	0.9553	1.21
UNSW-NB15	RFE@10	0.8891 ± 0.0008	+0.0007	0.9792	1.29
CIC-IDS2017	All features (77)	0.9983 ± 0.0001	+0.0017	0.9998	1.75
CIC-IDS2017	Consensus@10	0.9965 ± 0.0001	—	0.9993	0.40
CIC-IDS2017	Consensus@15	0.9975 ± 0.0001	+0.0010	0.9996	0.54
CIC-IDS2017	SHAP-only@10	0.9967 ± 0.0001	+0.0002	0.9993	0.42
CIC-IDS2017	Mutual info@10	$0.9920 \pm 0.0001 \dagger$	-0.0046	0.9986	0.46
CIC-IDS2017	Chi-square@10	$0.9570 \pm 0.0003 \dagger$	-0.0396	0.9913	0.45
CIC-IDS2017	RFE@10	0.9980 ± 0.0002	+0.0015	0.9998	0.48

4.6. Explanation cost

The explanation stage has costs of its own, and at operational scale they matter. TreeSHAP explained 2000 flows of the LightGBM models in 5.1-6.2 s (about 3 ms per flow), and the fast tree approximation explained 1000 flows of the RF models in 1.1-1.3 s. LIME required roughly 20-30 ms per explained flow across models and datasets, and dataset-level permutation importance completed in 2-21 s depending on model and dimensionality. Exact kernel-based SHAP for the MLP was excluded: its sampling complexity grows with feature count such that explaining even the 50-flow LIME sample

would exceed the entire training budget of the detector, an imbalance that practitioners should weigh when choosing explainers for line-rate systems.

5. Discussion

Implications for practice. The disagreement we measured (ECS between 0.213 and 0.340) means that single-explainer XAI reports in intrusion detection can be sensitive to the choice of explanation method: the specific explanation shown to an analyst depends as much on the choice of explainer as on the underlying detector. We recommend that explainability claims for IDS products and papers be accompanied by a cross-method agreement statement, for which ECS provides a simple, reproducible instrument; that alert-level explanations display only features endorsed by multiple methods, or flag single-method features as tentative; and that feature-selection decisions be based on consensus rankings rather than a single importance vector, since the latter embeds one method's biases into the deployed system permanently.

Why consensus helps. The NSL-KDD improvement is the result worth dwelling on: the full 122-feature model is measurably worse than the 10-feature consensus model on the distribution-shifted KDDTest+ partition. Features that only one pipeline ranks highly may be more sensitive to the selected explanation pipeline, particularly under LIME's sampling locality, permutation effects in the presence of correlated inputs, or SHAP's treatment of one-hot groups; demanding cross-method and cross-model support reduces the influence of features whose importance is sensitive to a particular explanation pipeline, and what remains is a compact and more stable feature subset under the evaluated conditions. The mechanism can be viewed as an ensemble-style aggregation of explanation rankings rather than prediction outputs.

Against standard selectors. The baseline comparison of Section 4.5 sharpens this interpretation. Filter methods such as mutual information and chi-square score features one at a time against the training labels, so they inherit exactly the training-distribution biases that distribution shift punishes; chi-square's collapse on NSL-KDD is the extreme case. Wrapper methods such as RFE optimize training-side performance of one model and transfer its idiosyncrasies to the selected set. The consensus differs from both in kind: it aggregates evidence across explanation mechanisms and model families, and features that survive that aggregation are the ones whose importance is a property of the data rather than of any single pipeline. That is why it is the only selector in our grid that beats the full feature set where generalization is hardest, while never falling meaningfully behind where in-distribution accuracy is saturated.

Limitations. First, the study covers binary detection; per-attack-class explanation agreement may behave differently and multiclass consensus is a natural extension. Second, CIC-IDS2017 is used through the published sample of Ref. [31] rather than the 2.8-million-flow original, and the UNSW-NB15 public partitions carry known label and encoding idiosyncrasies; both choices favor reproducibility over scale. Third, TreeSHAP for RF uses the fast path-dependent approximation, and LIME aggregates 50 local explanations per model; larger explanation samples would tighten the rankings, although the low agreement we report would, if anything, only become more precise. Fourth, the explanation grid is computed once with a fixed seed to keep the full grid within a strict compute budget; detection-side conclusions, however, are validated over five stratified training subsamples with paired significance tests (Section 4.5), and the consistency of the pattern across three independent datasets provides further evidence of robustness. Fifth, explanation faithfulness itself is not evaluated against ground truth, which no public IDS benchmark currently provides; agreement is a necessary but not sufficient condition for trustworthy explanations, and adversarial manipulation of explainers [15] remains an open threat model.

6. Conclusions

This paper measured, across a full grid of three attribution methods, three detector families, and three public benchmarks, how much post hoc explanations of network intrusion detectors actually agree, and it proposed a consensus framework that uses the observed disagreement to guide feature selection and produce more compact detectors. Explanation agreement is low, with Explanation Consensus Scores between 0.213 and 0.340, and the choice of explanation method changes the reported evidence about as much as changing the model family. Borda-count fusion of the eight explanation pipelines yields a consensus feature ranking that, with only ten features, improves F1-score on NSL-KDD from 0.766 to 0.799 over the full 122-feature detector, achieves performance close to or better than SHAP-only selection across the three benchmarks, outperforms mutual information, chi-square, and RFE baselines under distribution shift with statistically significant margins, reduces training time by approximately half, and keeps inference below 7 microseconds per flow. Future work includes multiclass and per-attack consensus, drift-aware consensus maintenance on streaming traffic, faithfulness benchmarks with synthetic ground truth, and integrating consensus scores directly into analyst-facing alert interfaces.

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Data availability statement

All datasets used in this study (NSL-KDD, UNSW-NB15, and CIC-IDS2017) are publicly available; the complete experimental code and result files are available from the corresponding author upon reasonable request.

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References

- [1] Buczak AL, Guven E. A survey of data mining and machine learning methods for cyber security intrusion detection. *IEEE Commun Surv Tutor*. 2016;18(2):1153–1176.
- [2] Khraisat A, Gondal I, Vamplew P, Kamruzzaman J. Survey of intrusion detection systems: techniques, datasets and challenges. *Cybersecurity*. 2019;2(1):20.
- [3] Sommer R, Paxson V. Outside the closed world: on using machine learning for network intrusion detection. In: *Proceedings of the 2010 IEEE Symposium on Security and Privacy*; 2010; Oakland, CA, USA. p. 305–316.
- [4] Vinayakumar R, Alazab M, Soman KP, Poornachandran P, Al-Nemrat A, Venkatraman S. Deep learning approach for intelligent intrusion detection system. *IEEE Access*. 2019;7:41525–41550.
- [5] Shone N, Ngoc TN, Phai VD, Shi Q. A deep learning approach to network intrusion detection. *IEEE Trans Emerg Top Comput Intell*. 2018;2(1):41–50.
- [6] Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat Mach Intell*. 2019;1(5):206–215.

- [7] Adadi A, Berrada M. Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access*. 2018;6:52138–52160.
- [8] Barredo Arrieta A, Diaz-Rodriguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, et al. Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf Fusion*. 2020;58:82–115.
- [9] Gunning D, Stefik M, Choi J, Miller T, Stumpf S, Yang GZ. XAI—explainable artificial intelligence. *Sci Robot*. 2019;4(37):eaay7120.
- [10] Neupane S, Ables J, Anderson W, Mittal S, Rahimi S, Banicescu I, et al. Explainable intrusion detection systems (X-IDS): a survey of current methods, challenges, and opportunities. *IEEE Access*. 2022;10:112392–112415.
- [11] Ribeiro MT, Singh S, Guestrin C. “Why should I trust you?” Explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*; 2016; San Francisco, CA, USA. p. 1135–1144.
- [12] Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: *Proceedings of the 31st International Conference on Neural Information Processing Systems*; 2017; Long Beach, CA, USA. p. 4765–4774.
- [13] Fisher A, Rudin C, Dominici F. All models are wrong, but many are useful: learning a variable’s importance by studying an entire class of prediction models simultaneously. *J Mach Learn Res*. 2019;20(177):1–81.
- [14] Krishna S, Han T, Gu A, Pombra J, Jabbari S, Wu S, et al. The disagreement problem in explainable machine learning: a practitioner’s perspective. *arXiv preprint arXiv:2202.01602*. 2022.
- [15] Slack D, Hilgard S, Jia E, Singh S, Lakkaraju H. Fooling LIME and SHAP: adversarial attacks on post hoc explanation methods. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*; 2020; New York, NY, USA. p. 180–186.
- [16] Wang M, Zheng K, Yang Y, Wang X. An explainable machine learning framework for intrusion detection systems. *IEEE Access*. 2020;8:73127–73141.
- [17] Warnecke A, Arp D, Wressnegger C, Rieck K. Evaluating explanation methods for deep learning in security. In: *Proceedings of the 2020 IEEE European Symposium on Security and Privacy*; 2020; Genoa, Italy. p. 158–174.
- [18] Arreche O, Guntur TR, Roberts JW, Abdallah M. E-XAI: evaluating black-box explainable AI frameworks for network intrusion detection. *IEEE Access*. 2024;12:23954–23988.
- [19] Gaspar D, Silva P, Silva C. Explainable AI for intrusion detection systems: LIME and SHAP applicability on multi-layer perceptron. *IEEE Access*. 2024;12:30164–30175.
- [20] Hermosilla P, Berrios S, Allende-Cid H. Explainable AI for forensic analysis: a comparative study of SHAP and LIME in intrusion detection models. *Appl Sci*. 2025;15(13):7329.
- [21] Zebin T, Rezvy S, Luo Y. An explainable AI-based intrusion detection system for DNS over HTTPS (DoH) attacks. *IEEE Trans Inf Forensics Secur*. 2022;17:2339–2349.
- [22] Mane S, Rao D. Explaining network intrusion detection system using explainable AI framework. *arXiv preprint arXiv:2103.07110*. 2021.
- [23] Capuano N, Fenza G, Loia V, Stanzione C. Explainable artificial intelligence in cybersecurity: a survey. *IEEE Access*. 2022;10:93575–93600.
- [24] Guidotti R, Monreale A, Ruggieri S, Turini F, Giannotti F, Pedreschi D. A survey of methods for explaining black box models. *ACM Comput Surv*. 2018;51(5):93.
- [25] Doshi-Velez F, Kim B. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*. 2017.
- [26] Molnar C. *Interpretable machine learning: a guide for making black box models explainable*. 2nd ed. Munich (Germany): Independently published; 2022.
- [27] Dwivedi R, Dave D, Naik H, Singhal S, Omer R, Patel P, et al. Explainable AI (XAI): core ideas, techniques, and solutions. *ACM Comput Surv*. 2023;55(9):194.
- [28] Tavallaee M, Bagheri E, Lu W, Ghorbani AA. A detailed analysis of the KDD CUP 99 data set. In: *Proceedings of the 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*; 2009; Ottawa, Canada. p. 1–6.

- [29] Moustafa N, Slay J. UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In: Proceedings of the 2015 Military Communications and Information Systems Conference; 2015; Canberra, Australia. p. 1–6.
- [30] Sharafaldin I, Lashkari AH, Ghorbani AA. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In: Proceedings of the 4th International Conference on Information Systems Security and Privacy; 2018; Funchal, Portugal. p. 108–116.
- [31] Yang L, Moubayed A, Shami A. MTH-IDS: a multitiered hybrid intrusion detection system for internet of vehicles. *IEEE Internet Things J.* 2022;9(1):616–632.
- [32] Breiman L. Random forests. *Mach Learn.* 2001;45(1):5–32.
- [33] Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: a highly efficient gradient boosting decision tree. In: Proceedings of the 31st International Conference on Neural Information Processing Systems; 2017; Long Beach, CA, USA. p. 3146–3154.
- [34] Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, et al. From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell.* 2020;2(1):56–67.
- [35] Kendall MG. A new measure of rank correlation. *Biometrika.* 1938;30(1/2):81–93.